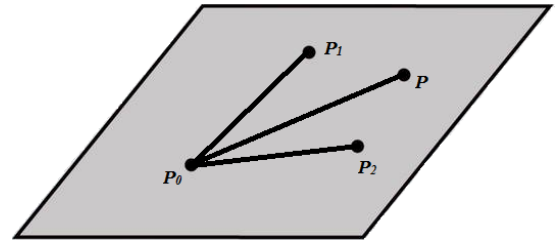


MTH 202 Chapter 1: Three-Dimensional Space & Vectors

1.6 Planes in 3-Space

We now move to thinking about planes, and we ask ourselves what information do we need in order to determine an equation of a plane?



Definition: We say that a vector is normal to a plane if it is perpendicular to the plane.

Terminology: If $\mathbf{r}_0$ is the position vector of a point on a plane and $\mathbf{n}$ is a vector normal to the plane, then $\mathbf{n} \cdot (\mathbf{r} - \mathbf{r}_0) = 0$ is called the vector equation of the plane.

Note: The vector equation of a plane can be used beyond three dimensions, and it can be used in two dimensions to yield the equation of a line! In this class, planes in four or more dimensions will be known as hyperplanes.

If we return our focus to 3-space and suppose that

Terminology: The equation $a(x - x_0) + b(y - y_0) + c(z - z_0) = 0$ is known as the point-normal form of the equation of a plane through (x_0, y_0, z_0) with normal vector $\mathbf{n} = \langle a, b, c \rangle$.

Theorem: If a , b , c , and d are constants (with a , b , and c not *all* zero), then the graph of the equation

$$ax + by + cz = d$$

is a plane that has the vector $\mathbf{n} = \langle a, b, c \rangle$ as a normal vector.

Note: Alternative approaches to deriving this equation are shown in the notes in Blackboard.

Example 1: Find an equation (in the form $ax + by + cz = d$) of the plane through the point $(2, 4, -1)$ with normal vector $\langle 2, 3, 4 \rangle$. Also sketch the portion of this plane which is in the first octant.

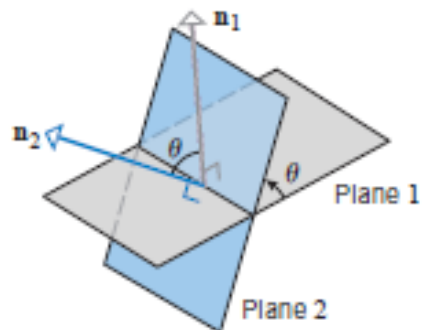
Example 2: Find an equation of the plane that passes through the points $A: (1, 3, 3)$, $B: (3, -1, 5)$, and $C: (5, 2, -1)$.

Definition: Two planes are parallel if their normal vectors are parallel. If two planes are not parallel, then they intersect in a straight line and the angle between the two planes is defined to be the acute or right angle between their normal vectors (one must choose the normal vectors so that their dot product is nonnegative).

Example 3: Consider the planes given by $2x - 2y + z = 4$ and $x + 4y + z = 1$.

a) Determine whether the planes parallel.

b) Find the angle (rounded to the nearest hundredth) between the planes.



c) Write a parametric equation for the line of intersection for the planes.

Example 4: Find the exact distance between the point $A: (-1, 1, 2)$ and the plane given by: $x + 4y + z = 1$.

Note: A way of generalizing this process is shown in the notes on Blackboard.